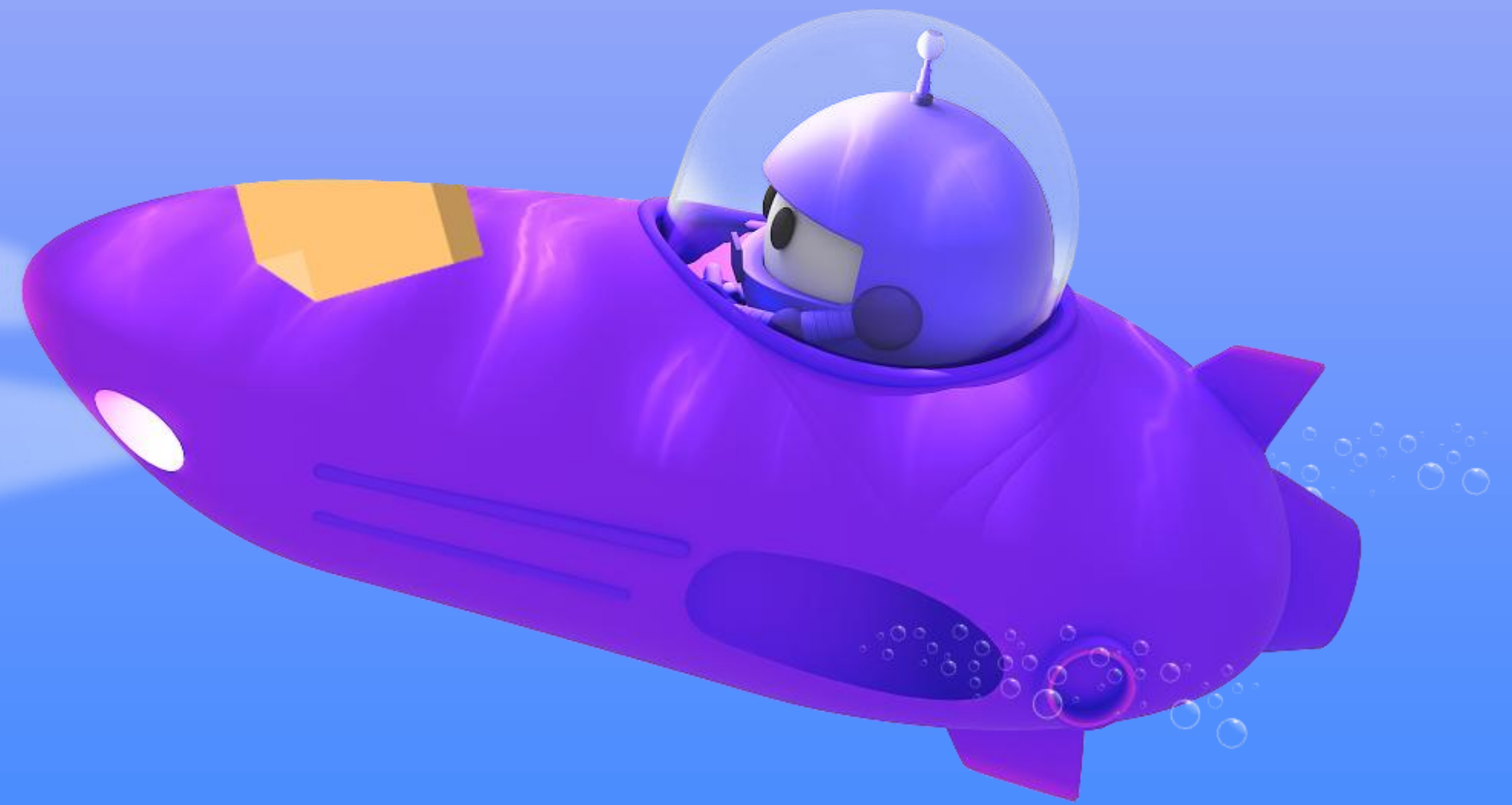


Take a Note of UI with .NET 10

Peter Bull

Microsoft MVP – .NET & Windows Development



About

Peter Bull, Microsoft MVP .NET & Windows Development

- Live in Gateshead & Senior Software Engineer – klipboard.com
- Creation, Packages & Learning – comentsys.com
- Posts, Articles & Podcast – rogueplanetoid.com
- Tutorials, Talks & Workshops – tutorialr.com
- LinkedIn – linkedin.com/in/rogueplanetoid
- GitHub – github.com/rogueplanetoid



.NET

Free, open-source, cross-platform framework for building modern apps & powerful cloud services

.NET 10

Runtime & Libraries

.NET runtime improvements, new library APIs & support for file-based apps

C# 14

Extension Members

Syntax for extension members enable extension properties to be defined

Native

Windows Forms & WPF

Windows Forms enhancements & fluent style changes or improvements to WPF

Fields

Field-backed properties

Enables property accessor body without need to declare an explicit backing field

Web

ASP.NET Core & Blazor

ASP.NET Core features improved form validation, Blazor reconnection UI component & state persistence

Assign

Null-conditional assignment

Now supports null-conditional member access operators that can be used on left side of an assignment



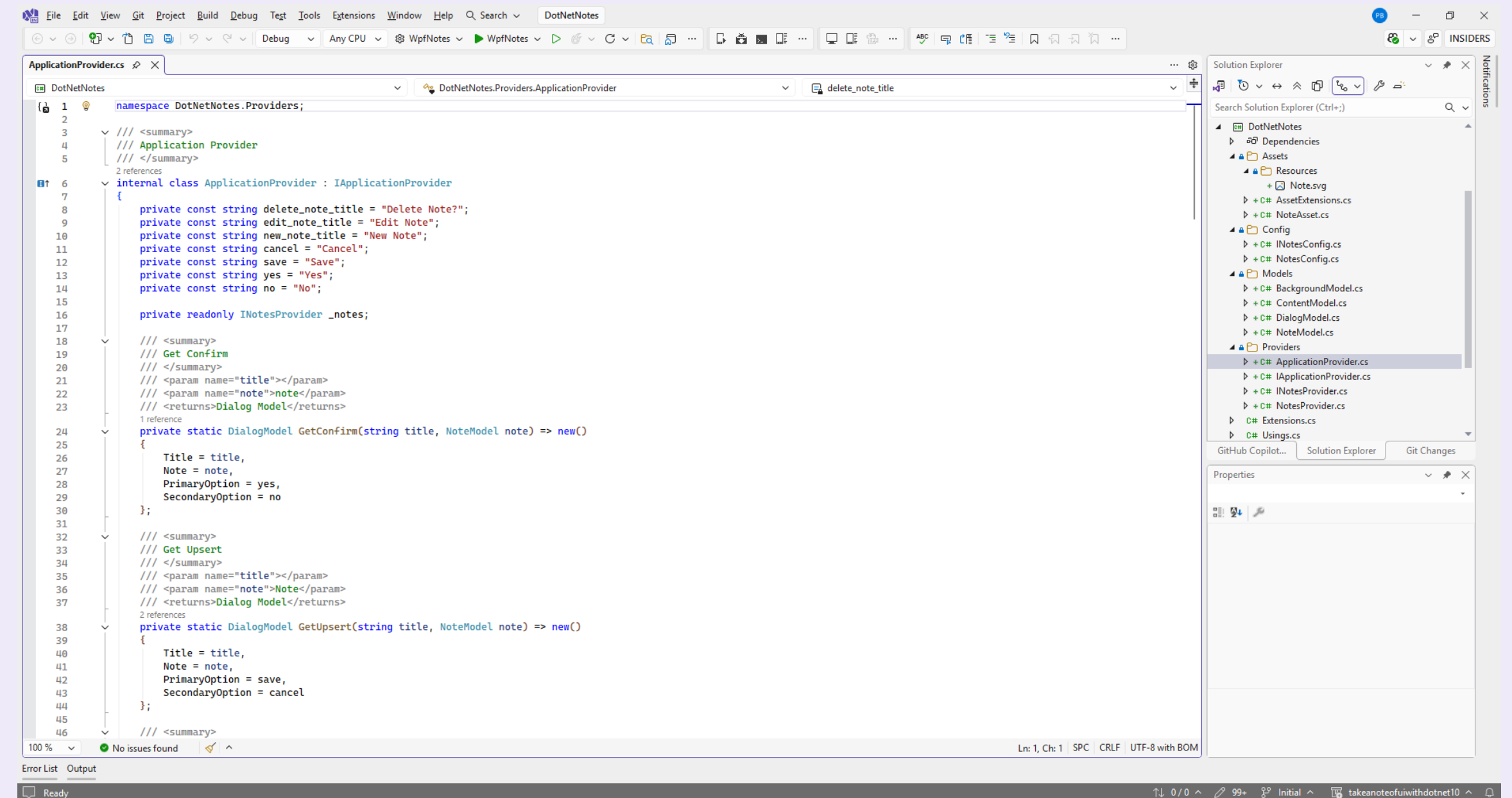
.NET

dot.net

Visual Studio 2026

Stay in your flow & level up with AI

- Visual Studio 2026 builds on what developers use to make it more intuitive to help developers build something new or scale existing solutions
- Visual Studio 2026 features a refreshed & modern Fluent Design with updated icons, improvements to user interface & increased performance
- Visual Studio 2026 deeply integrates with GitHub Copilot to understand your work with context across your solution, repository or tools
- Visual Studio 2026 offers compatibility for existing workloads & extensions from Visual Studio 2022 providing future-ready IDE for .NET 10

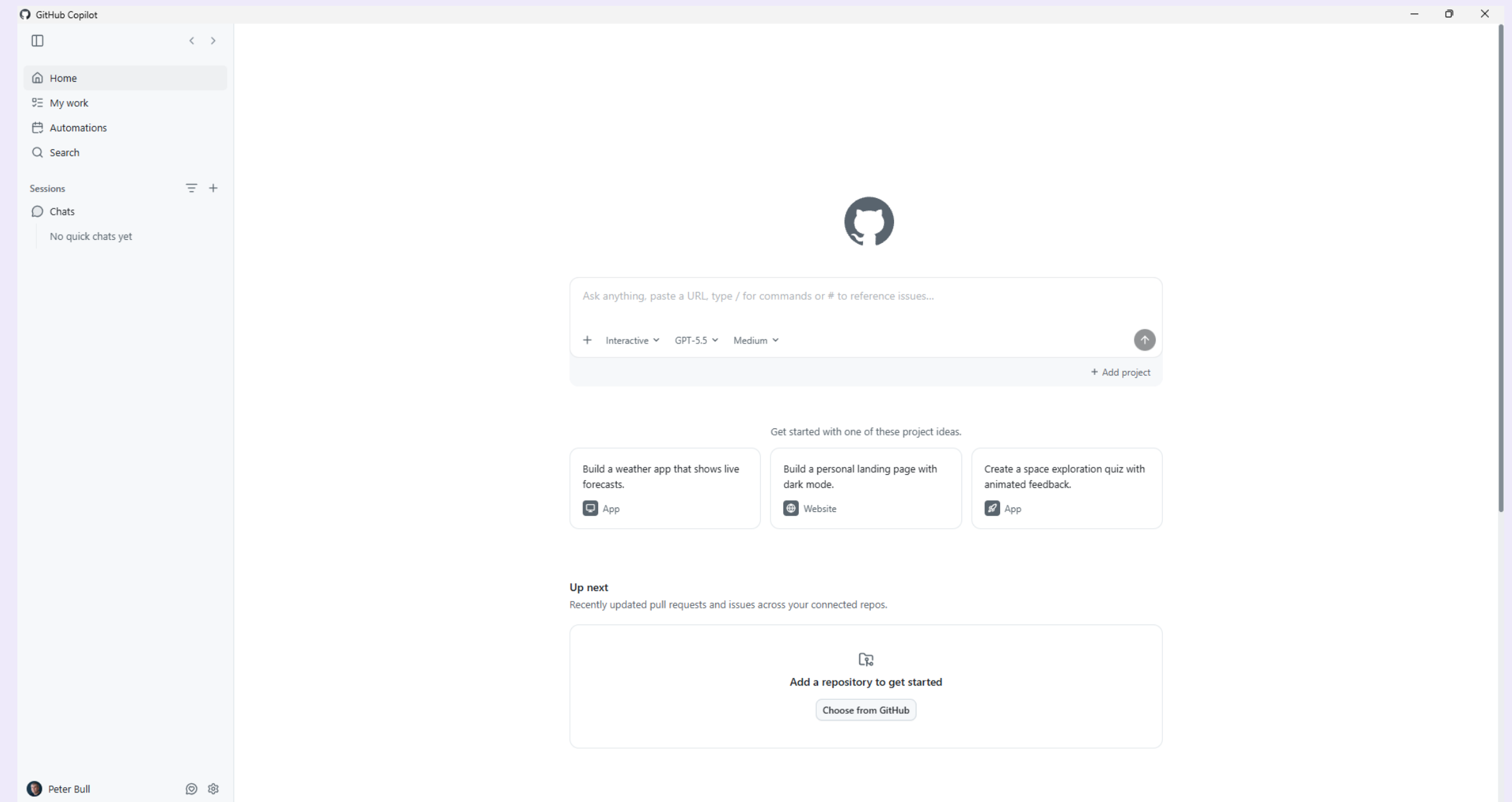


visualstudio.com

GitHub Copilot

AI that builds with you

- GitHub Copilot helps transform the developer experience to help focus on problem solving
- GitHub Copilot can analyse code, propose edits, create functionality, fix bugs or mistakes
- GitHub Copilot can perform actions by leveraging a wide range of tools using MCP servers
- GitHub Copilot has modes of ask to control changes or agent to apply changes



github.com/app

Take a Note of UI with .NET 10

.NET first-party user interface technologies

UI in .NET

A to (almost) Z of first-party UI in .NET
& how GitHub Copilot helped



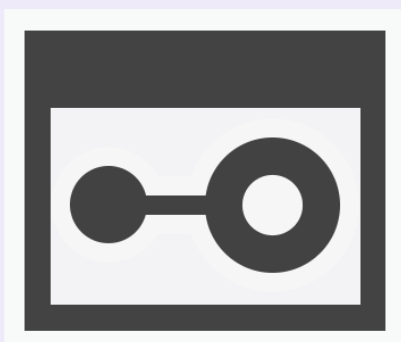
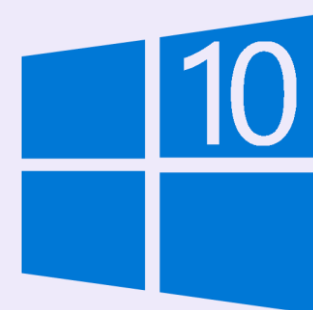
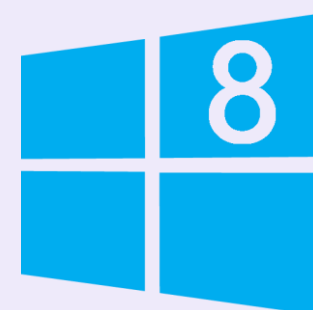
.NET

ASP.NET Core Web API

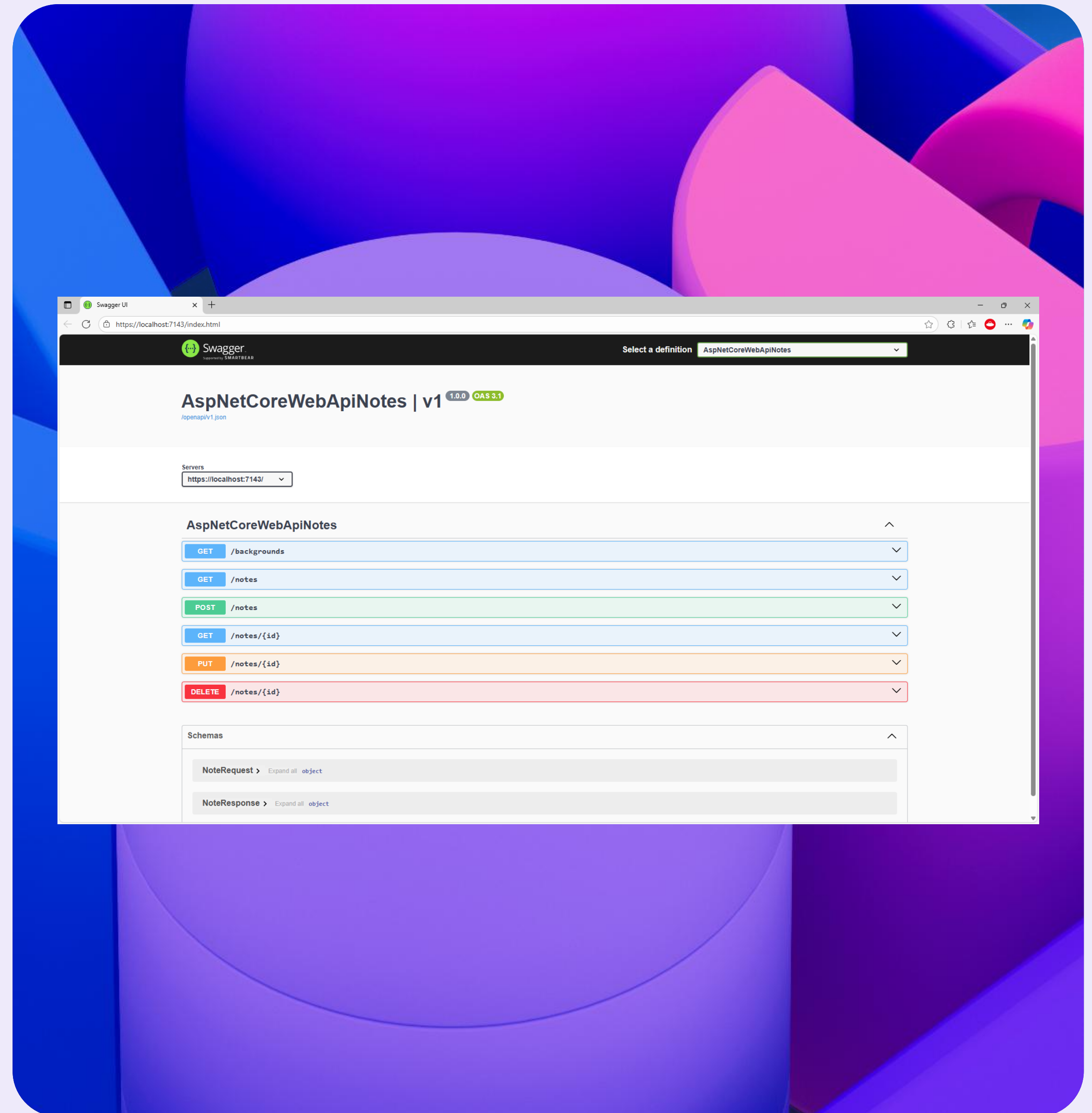
RESTful Web APIs using ASP.NET Core controllers or minimal APIs



- Create suitable API endpoints from a Notes application
- Create an allowed background attribute
- Create a .http file for my API



macOS



ASP.NET Core Web App

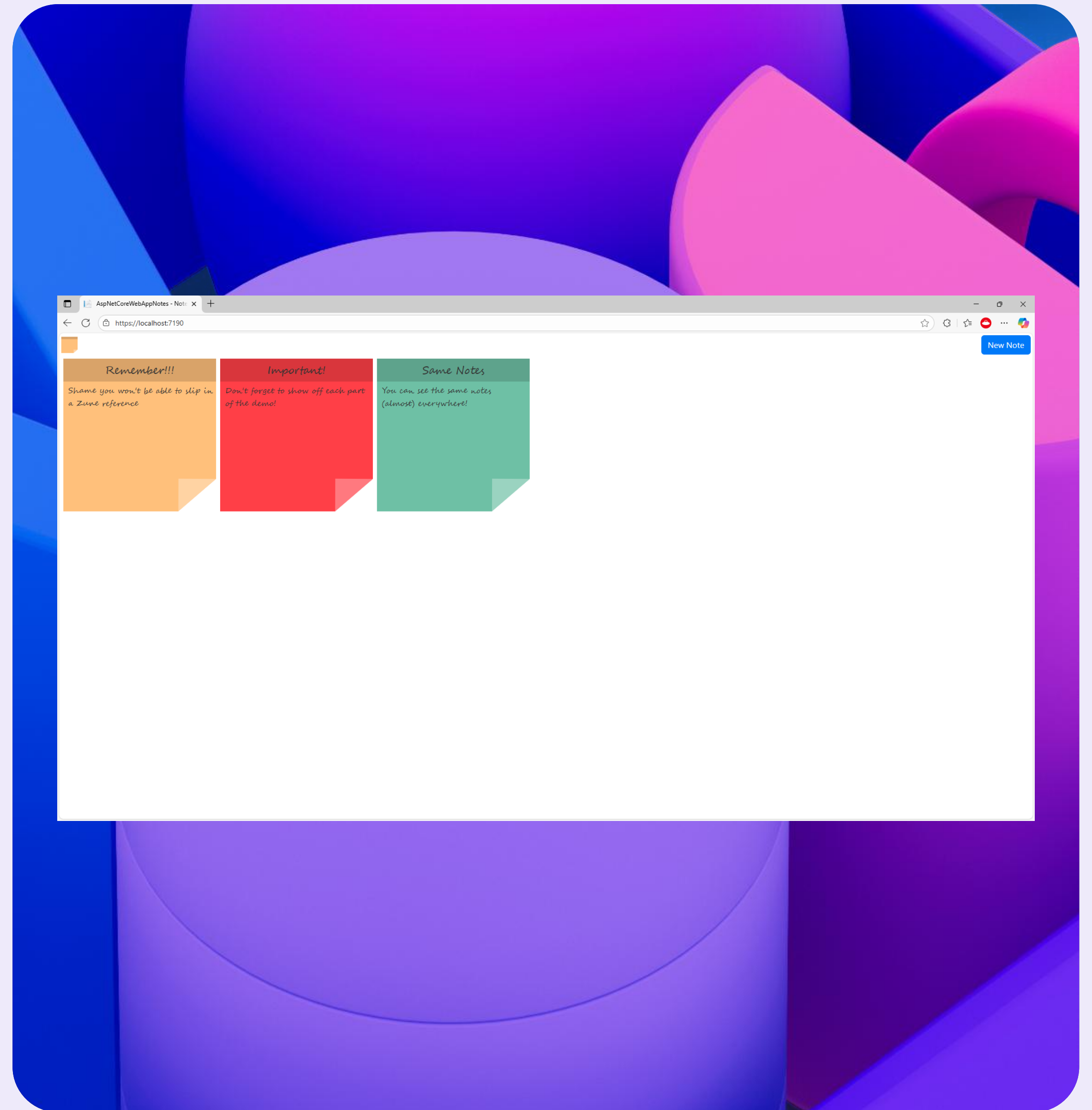
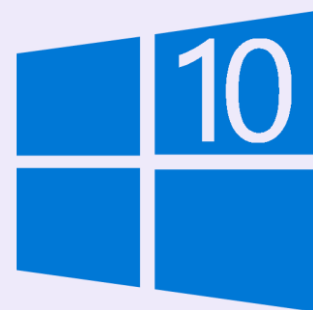
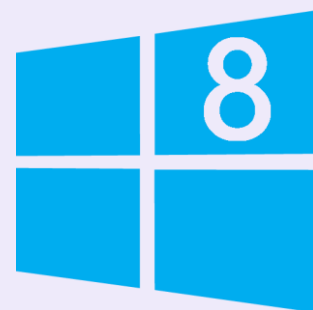
ASP.NET Core application using Model-View-Controller or Razor Pages



- Add code to keep modal open if model state is not valid



macOS



Blazor Web App

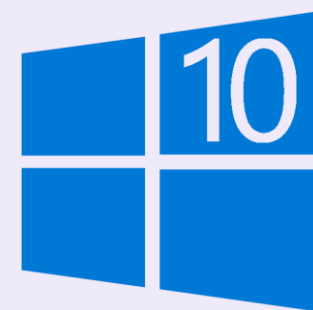
Blazor web app supporting server-side rendering and client interactivity



- Update modal to support edit and new



macOS

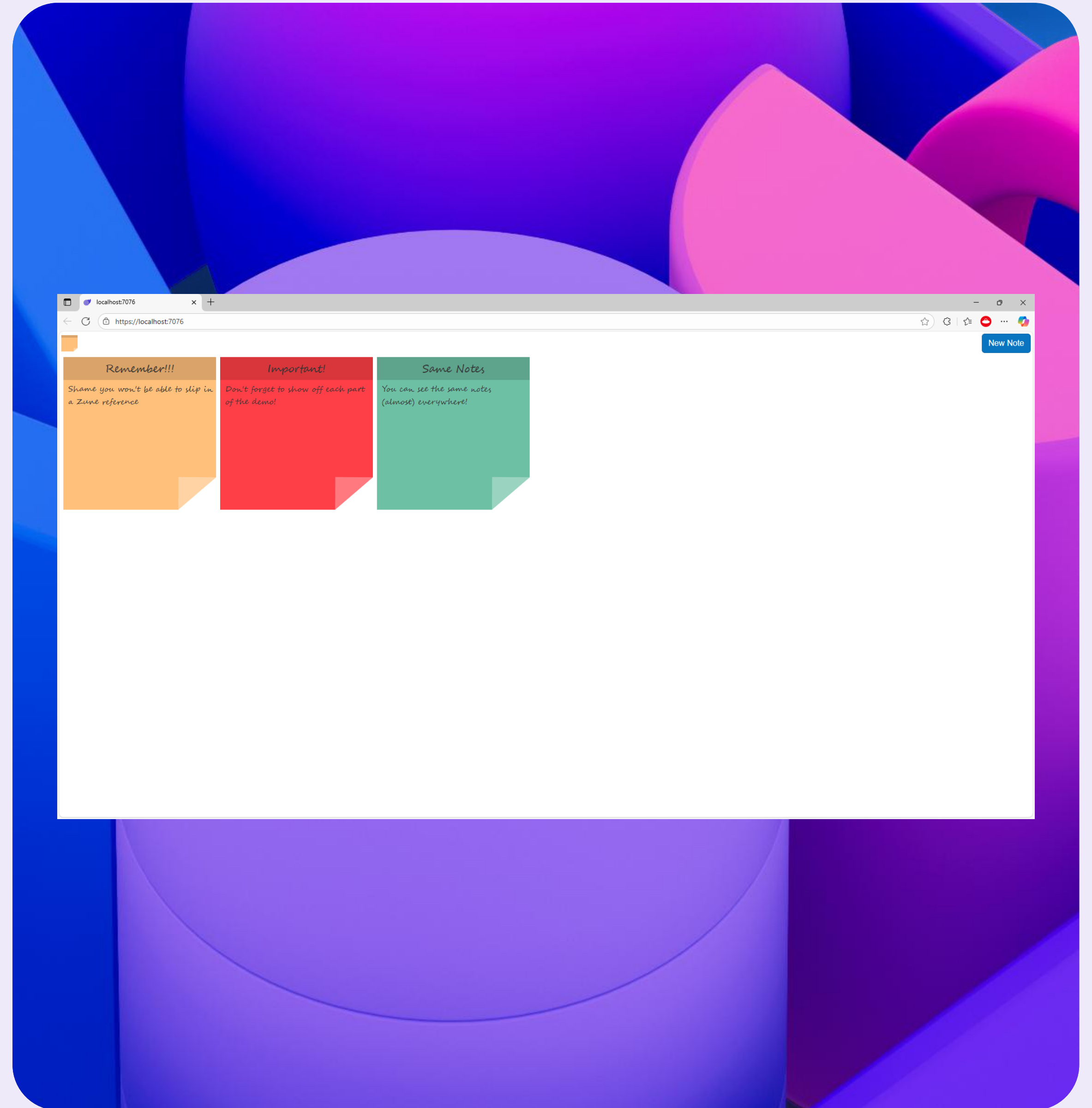


Blazor WebAssembly App

Blazor WebAssembly for web apps with rich dynamic user interfaces



- Create local storage-based based notes provider

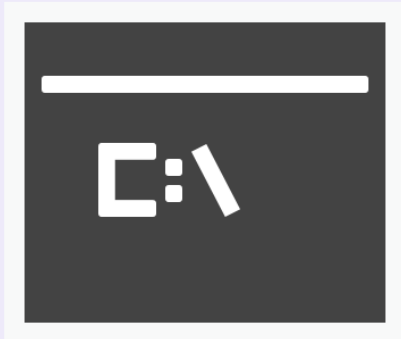
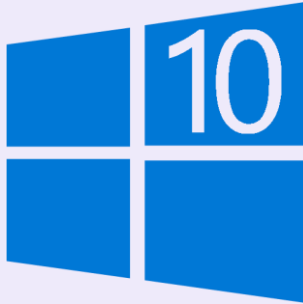
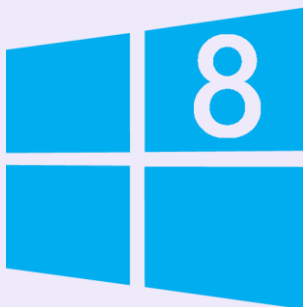


Console App

Command-line application for Linux, macOS & Windows



- Create a Console application based upon Notes application
- Create command-line argument parser
- Update with interactive mode if no arguments provided
- Update to accept valid backgrounds
- Update to quit interactive mode with Esc
- Update to confirm before deletion

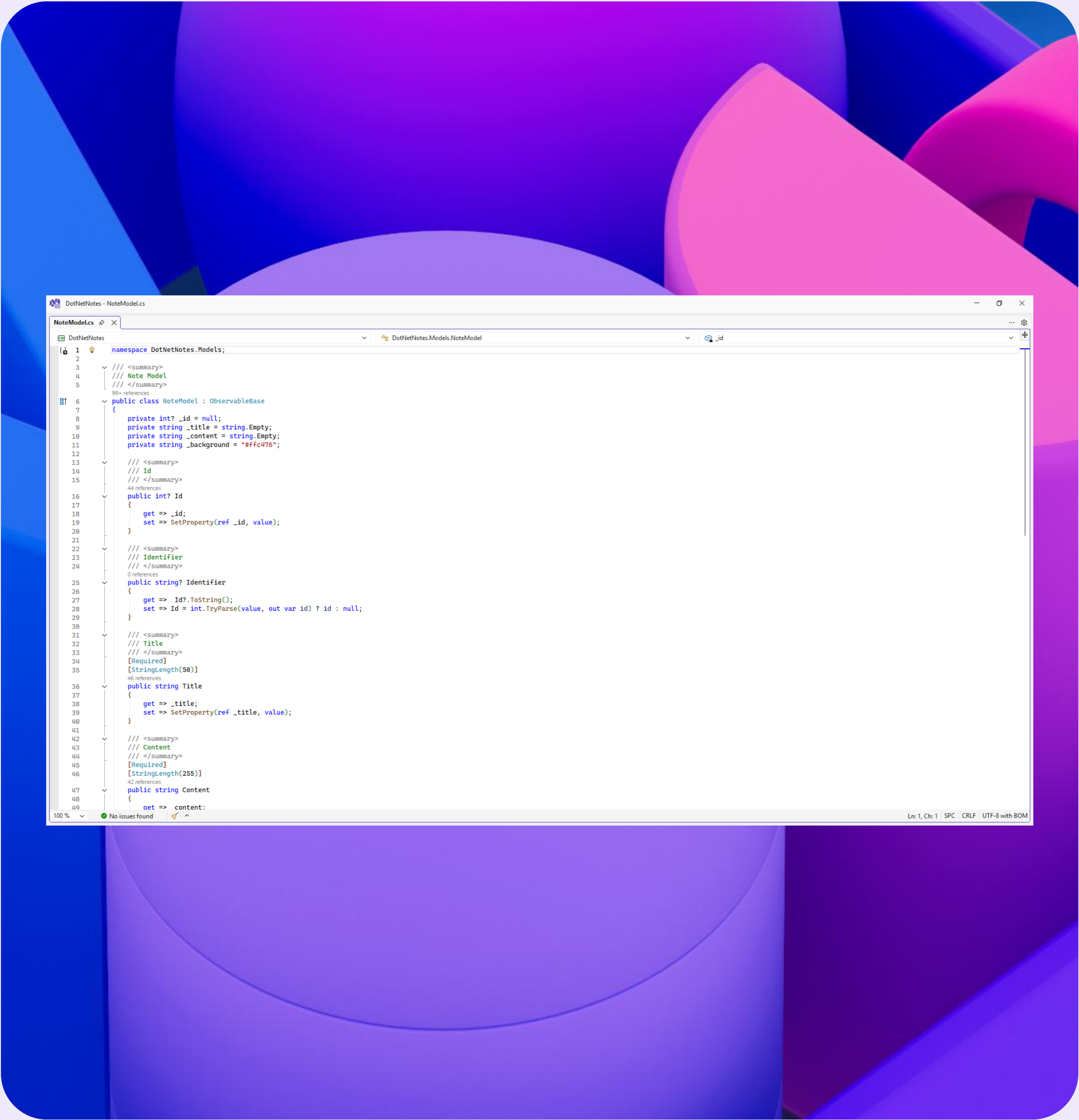


Class Library

Reusable collection of classes, interfaces & methods



- Update methods with required attributes



Local MCP Server Console App

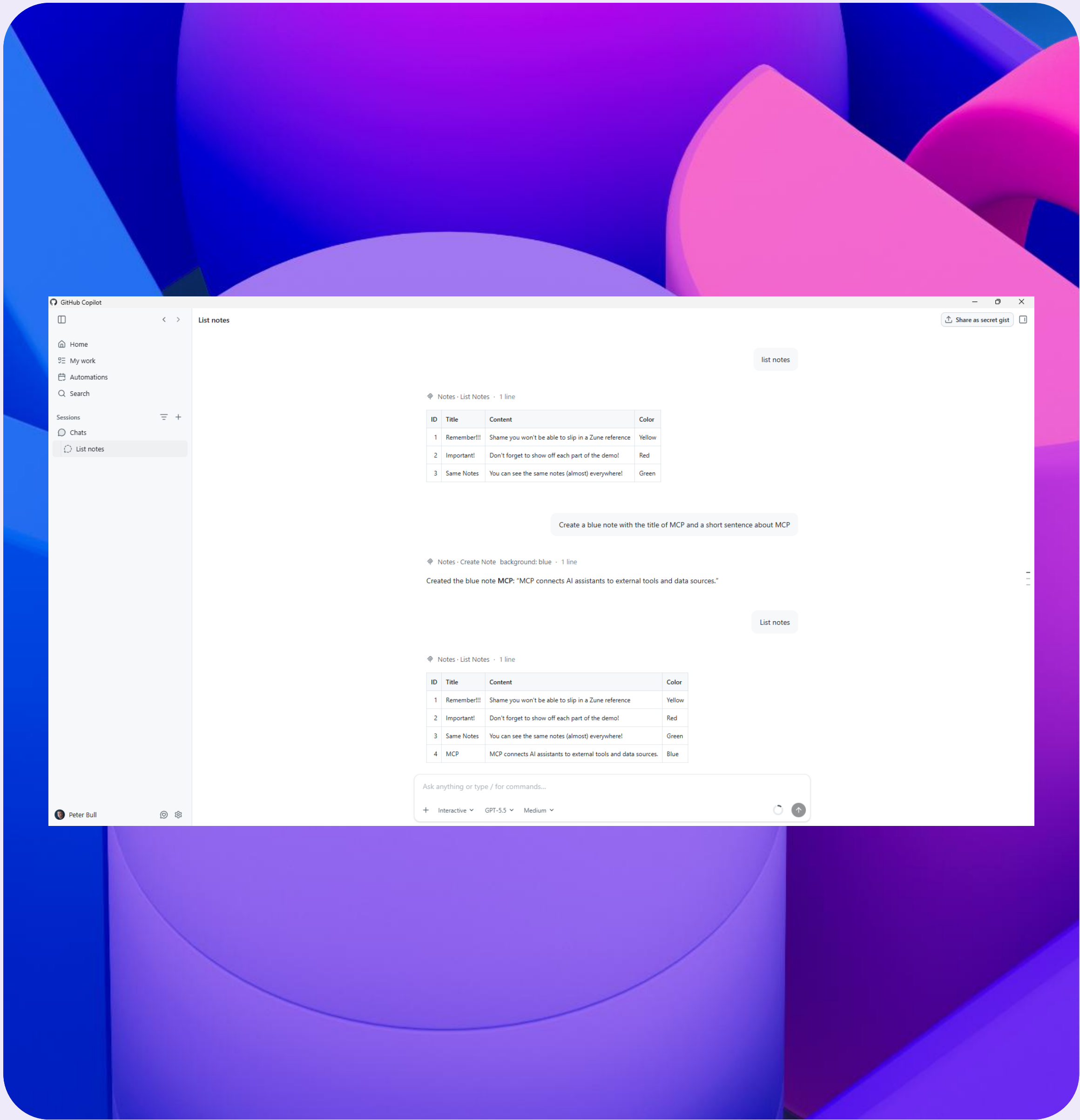
Model Context Protocol server command-line application



- Create an MCP tool for my Notes application



macOS

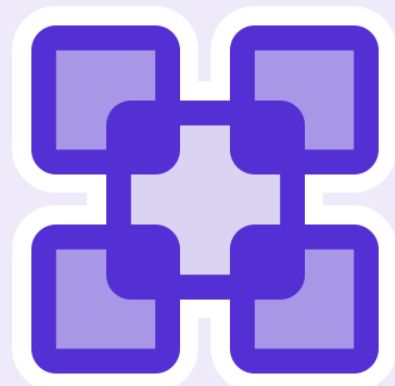


.NET MAUI App

Application for iOS, Android, Mac Catalyst & WinUI



- Update Note Control to be scalable
- Update Popup with latest changes in .NET MAUI Community Toolkit



Apple iOS



Apple macOS



UWP App

Universal Windows Platform app using .NET & Native AoT



- Create a UWP app based upon Notes application
- Create Relay Command, View Model and Converters for Note and Content Model
- Create a HTML Colour to Windows Colour Converter
- Update Class Library methods with required attributes for AoT

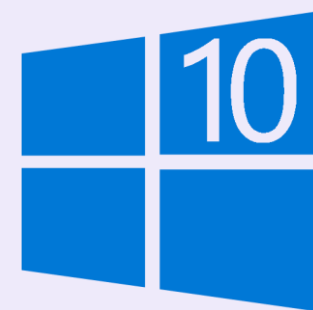
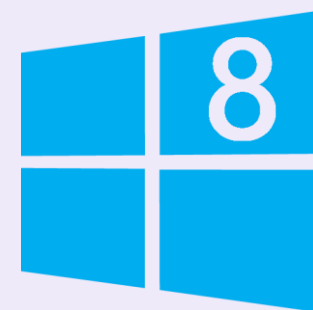


Windows Forms App

WinForms App



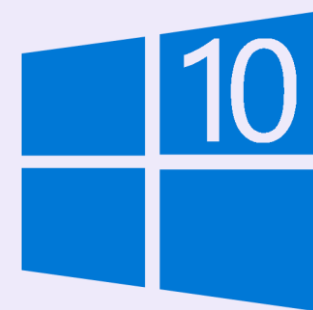
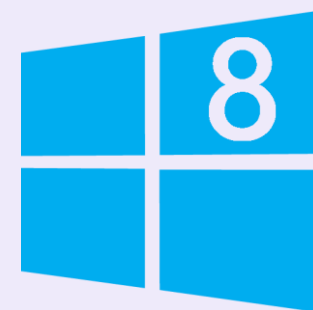
- Create a Windows Form app based upon Notes application
- Create a Note Control based upon the Note Control from Notes application
- Update Note Control so that corner is on the bottom right instead of the top right



Notification Area or Task Tray icon from Windows Forms App



- Create a Notification Area implementation based upon Notes application
- Update implementation with an option to exit the application
- Update implementation to be loaded when Windows Forms application is closed and hidden when Windows Forms application is loaded

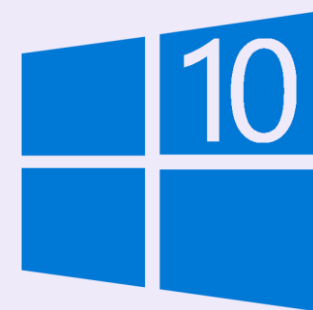
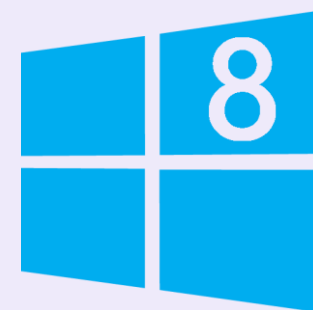
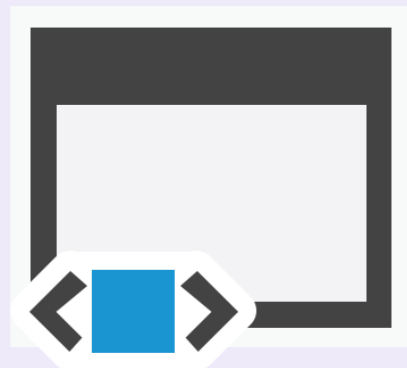


WPF Application

Windows Presentation Foundation app



- Create a Note Control based upon the Note Control from Notes application



WinUI App

Windows App SDK & MSIX side-loading or distribution via Microsoft Store



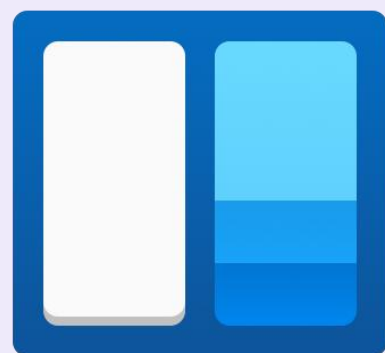
- Create a Console App, UWP App along with Windows Form App based upon this Notes application
- Create Note Control for WPF app based on Note Control from this Notes application
- Add Titlebar from Windows App SDK 1.8



Widget Provider from WinUI App using Windows App SDK



- Create Adaptive Cards for Configuration with New and Refresh at the top, a list of three Notes with Edit and Delete with Paging at the bottom
- Create a Confirmation Adaptive Card based on Confirmation from WinUI application
- Create an Adaptive Card for creating or editing Notes for Notes application
- Create a service for Configuration which includes support for paging of notes



NuGet
Create .NET apps faster



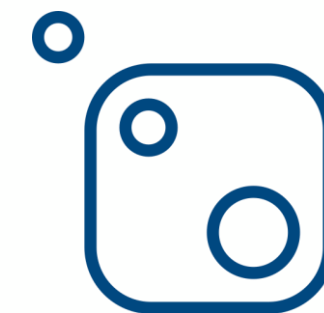
Comentsys.Toolkit

Toolkit for .NET supporting
SVG-based assets & Data
Binding for XAML apps



Comentsys.Toolkit. WindowsAppSdk

Toolkit for Windows App SDK
including support for Widget
Providers in apps



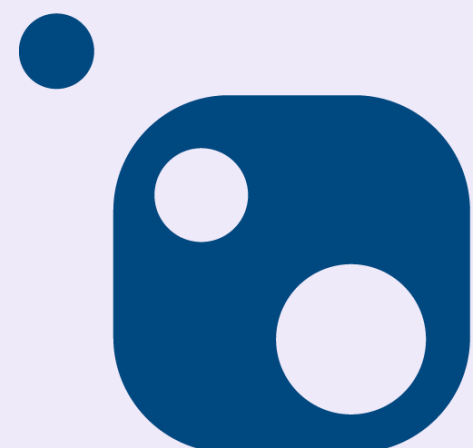
ModernWpfUI

Modern styles & features for
Windows Presentation
Foundation apps



CommunityToolkit. Maui

Features for .NET MAUI apps
including popup for
presenting information



nuget.org

DEMO

Note apps with .NET first-party UI

Take a Note of UI in .NET 10

A screenshot of the Visual Studio IDE showing the source code for the NotesEndpoints.cs file. The code is written in C# and defines the endpoints for a notes application. The Solution Explorer on the right shows the project structure, including the NotesEndpoints.cs file. The status bar at the bottom indicates "No issues found" and "Ln: 1, Ch: 1 SPC CRLF UTF-8 with BOM".

```
1 namespace AspNetCoreWebApiNotes.Endpoints;
2
3 /// <summary>
4 /// Notes Endpoints
5 /// </summary>
6 public static class NotesEndpoints
7 {
8     /// <summary>
9     /// Map Notes Endpoints
10    /// </summary>
11    /// <param name="endpoints">Endpoint Route Builder</param>
12    /// <returns>Endpoint Route Builder</returns>
13    public static IEndpointRouteBuilder MapEndpoints(this IEndpointRouteBuilder endpoints)
14    {
15        // List Backgrounds
16        endpoints.MapGet("/backgrounds", () =>
17            Results.Ok(BackgroundModel.Options.Select(background => new { background.Name, background.Background })))
18            .WithName("ListBackgrounds");
19        // Notes Group
20        var group = endpoints.MapGroup("/notes");
21        // List Note
22        group.MapGet("/", async (IApplicationProvider appProvider) =>
23        {
24            await appProvider.ListAsync();
25            return Results.Ok(appProvider.Content.Notes.Select(note => new NoteResponse(note)));
26        })
27        .WithName("ListNotes")
28        .Produces<IEnumerable<NoteResponse>>(StatusCodes.Status200OK);
29        // Get Note by Id
30        group.MapGet("/{id:int}", async (int id, IApplicationProvider appProvider) =>
31        {
32            var note = await appProvider.GetAsync(id);
33            return note is null ? Results.NotFound() : Results.Ok(new NoteResponse(note));
34        })
35        .WithName("GetNote")
36        .Produces<NoteResponse>(StatusCodes.Status200OK)
37        .Produces(StatusCodes.Status404NotFound);
38        // New Note
39        group.MapPost("/", async (NoteRequest request, IApplicationProvider appProvider) =>
40        {
41            if (request == null)
42                return Results.BadRequest("Create Note Request is required.");
43            var note = request.ToNoteModel();
44            var ok = await appProvider.NewAsync(note, true);
45            if (!ok || note.Id is null)
46                return Results.BadRequest("Unable to create note.");
47            return Results.Created($" /notes/{note.Id}", new NoteResponse(note));
48        })
49        .WithName("CreateNote")
50    }
51 }
```

Take a Note of UI with .NET 10

.NET third-party user interface technologies



Uno Platform

Build cross-platform .NET applications
fast along with AI powered visual tools

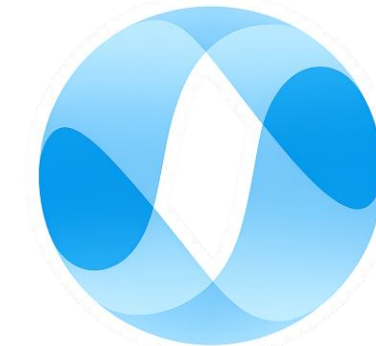
platform.uno



Avalonia UI

Build cross-platform desktop, mobile &
web apps with shared XAML & C#

avaloniaui.net



OpenSilver

Build modern .NET web apps in C# &
XAML effortlessly

opensilver.net

Take a Note of UI with .NET 10

Summary



01

Cloud & Web

- Create cloud & web apps using ASP.NET Core Web API / App
- Build Blazor Web apps for server or client using WebAssembly

02

Console & Class Library

- Craft console apps including local Model Context Protocol servers
- Contain common functionality for apps with Class Libraries

03

Mobile & Desktop

- Make cross-platform mobile & desktop apps using .NET MAUI
- Define Windows desktop apps with UWP, WinForms, WinUI, or WPF



Take a Note of UI with .NET 10 Questions

dot.net

visualstudio.com

github.com/app

nuget.org



github.com/RoguePlanetoid/takeanoteofuiwithdotnet10